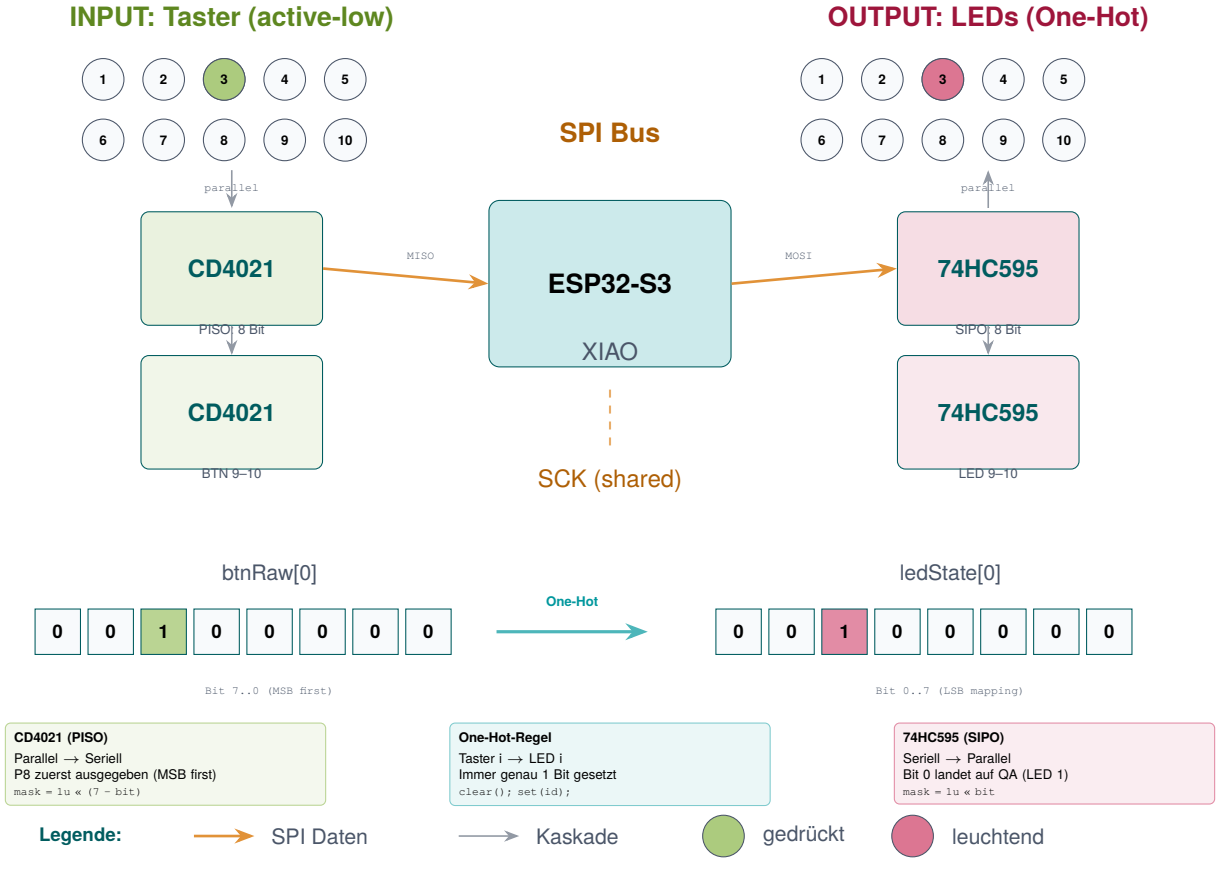
INPUT: Taster (active-low)
OUTPUT: LEDs (One-Hot)
1
2
3
4
5
6
7
8
9
10
parallel
SPI Bus
CD4021
MISO
ESP32-S3
XIAO
MOSI
74HC595
PISO 8 Bit
SIPO 8 Bit
CD4021
74HC595
BTN 9–10
SCK (shared)
LED 9–10
btnRaw[0]
One-Hot
ledState[0]
0
0
1
0
0
0
0
0
Bit 7..0 (MSB first)
Bit 0..7 (LSB mapping)
CD4021 (PISO)
Parallel → Seriell
P8 zuerst ausgegeben (MSB first)
mask = 1u « (7 - bit)
One-Hot-Regel
Taster i → LED i
Immer genau 1 Bit gesetzt
clear(); set(id);
74HC595 (SIPO)
Seriell → Parallel
Bit 0 landet auf QA (LED 1)
mask = 1u « bit
Legende:
SPI Daten
Kaskade
gedrückt
leuchtend